MATH2241 Introduction to Mathematical Analysis
Test 2
April 17, 2023 Name:
3:30 pm - 4:30 pm Student Number:

Attempt ALL Six Questions

Q1. Show that if f is continuous on $[a, b]$ and $f(x) \in [a, b]$ for every $x \in [a, b]$, then f has a fixed point, i.e., there exists an $x_0 \in [a, b]$ with $f(x_0) = x_0$.

Q2. Suppose that f is continuous on $[a, \infty)$. Show that if $\lim_{x\to\infty} f(x)$ exists then f is uniformly continuous on $[a, \infty)$.

Q3. Show that if f is one-to-one and continuous on $[a, b]$, then f^{-1} is continuous on $[m, M]$ where m and M are the minimum and the maximum values of f on $[a, b]$.

Q4. Show that if f' exists and is bounded on some interval I then f is uniformly continuous on I.

Q5. Show that if $a_k, b_k > 0$ for $k \in \mathbb{N}$,

$$\lim_{k\to\infty} \frac{b_k}{a_k} = 0 \quad \text{and} \quad \sum_{k=1}^{\infty} a_k \text{ converges},$$

then

$$\sum_{k=1}^{\infty} b_k \text{ converges}.$$

Also show that if $a_k, b_k > 0$ for $k \in \mathbb{N}$,

$$\lim_{k\to\infty} \frac{b_k}{a_k} = \infty \quad \text{and} \quad \sum_{k=1}^{\infty} a_k \text{ diverges},$$

then

$$\sum_{k=1}^{\infty} b_k \text{ diverges}.$$

Q6. Evaluate

$$\lim_{x\to 0^+} (\sin x)^x.$$

Q1. If $f(a) = a$ of $f(b) = b$, then it is valid. Assume $f(a) > a$ and $f(b) < b$. Define $g(x) = f(x) - x$ for every $x \in [a, b]$. Clearly $g(a) > 0$ and $g(b) < 0$, and g is continious on $[a, b]$ Now 0 is an intermediate value for g on $[a, b]$. Hence by the intermediate value theorem, there exists $x_0 \in (a, b)$ such that $g(x_0) = 0$, i.e., $f(x_0) = x_0$.

Q2. $\lim_{x\to\infty} f(x) = L$. Let $\epsilon > 0$ be given. There exists $M > a$ such that $x > M$, $|f(x) - L| < \epsilon/2$. f is continuous at $x = M$. There exists a $\delta_1 > 0$ such If $x \geq a$ and $|x - M| < \delta_1$, then $|f(x) - f(M)| < \epsilon/2$. Firstly, $f(x)$ is uniformly continuous on $[a, M]$. There exists a $\delta_2 > 0$ such that for $|x_1 - x_2| < \delta_2$ with $x_1, x_2 \in [a, M]$, $|f(x_1) - f(x_2)| < \epsilon$. Let $\delta = \min\{\delta_1, \delta_2\}$. Suppose $u, v \in [a, \infty)$ with $|u - v| < \delta$. (i) $u, v > M$, then $|f(u) - f(v)| \leq |f(u) - L| + |L - f(v)| < \epsilon/2 + \epsilon/2 = \epsilon$. (ii) $u, v \leq M$, because $|u - v| < \delta \leq \delta_2$, we have $|f(u) - f(v)| < \epsilon$. (iii) $u \leq M$ and $v > M$. Since $|u - v| < \delta$, we have $|u - M| < \delta \leq \delta_2$ and $|v - M| < \delta \leq \delta_1$. Thus $|f(u) - f(v)| \leq |f(u) - f(M)| + |M - f(v)| < \epsilon$.

Q3. The range of f is the closed, bounded interval $[m, M]$, and f^{-1} is well-defined one-to-one function on $[m, M]$. Let $y' \in [m, M]$. We would like to show that f^{-1} is continuous at y'. Let $\{y_n\}$ be any sequence in $[m, M]$ with $y_n \to y'$. We need to show that $f^{-1}(y_n) \to f^{-1}(y')$. Let $x' = f^{-1}(y')$ and $x_n = f^{-1}(y_n)$ for $n \in \mathbb{N}$. Suppose $\{x_n\}$ does not tend to x'. Then for some $\epsilon > 0$, $|x_n - x'| \geq \epsilon$ for infinitely many natural numbers n. We extract a subsequence $\{x_n^*\}$ of $\{x_n\}$ such that $|x_n^* - x'| \geq \epsilon$ for all n. $\{x_n^*\}$ is a bounded sequence, it has a convergent subsequence $\{\hat{x}_n\}$ converging to c in $[a, b]$. It is clear that $c \neq x'$. f is continuous at c, $f(\hat{x}_n) \to f(c)$. Since $\{f(\hat{x}_n)\}$ is a subsequence of $\{f(x_n)\} = \{y_n\} \to y' = f(x')$, it follows that $f(c) = f(x')$. It contradicts the assumption that f is one-to-one. Therefore, f^{-1} is continuous at y'.

Q4. $|f'(x)| \leq M$ for $x \in I$. For $x_1, x_2 \in I$ with $x_1 < x_2$, there exists $x_0 in (x_1, x_2)$ such that $|f'(x_0)| = |(f(x_2) - f(x_1))/(x_2 - x_1)| \leq M$, i.e., $|f(x_1) - f(x_2)| \leq M|x_2 - x_1|$ for $x_2, x_1 \in I$. Choose $\delta = \epsilon/M$.

Q5. $|b_k| < |a_k| r$ for $k \geq k_1$. $\{a_k\}$ is a Cauchy sequence implies that $\{b_k\}$ is a Cauchy sequence. Also $\lim_{k\to\infty} a_k/b_k = 0$. Suppose $\sum_k b_k$ converges, it implies that $\sum_k a_k$ converges. It is a contradiction.

Q6. $f(x) = e^x$ and and $f^{-1}(x) = lnx$ for $x > 0$. Both f and f^{-1} is continuous.

$$ln(\sin x)^x = xln(\sin x) = \frac{ln(\sin x)}{1/x}, \quad 0 < x < \pi.$$

$$\lim_{x\to 0^+} \frac{ln(\sin x)}{1/x} = \lim_{x\to 0^+} \frac{\cos x/\sin x}{-1/x^2} = 0.$$

The limit is 1.